



Recent advances to accelerate purification process development: a review with a focus on vaccines

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Recent advances to accelerate purification process development: a review with a focus on vaccines

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Abstract

Vaccination is yearly saving millions of lives from infectious diseases and provides social and economic benefits. The regulations are extremely high and likewise reflected in the process development time and costs. Reducing time-to-market is key for pharmaceutical companies, hence saving lives and money, and therefore the need raised for systematic, general and efficient process development strategies. High Throughput technologies tremendously increased the volume of process related data available and combined with statistical and mechanistic modeling new high throughput process development (HTPD) approaches evolved. Despite the tremendous variation between vaccine purification processes, platform processes for similar types of vaccines could aid to generally accelerate the process development.

This review provides an overview of downstream process development strategies and tools used within the (bio)pharmaceutical industry, focusing attention at (subunit) vaccines purification processes. Subsequent high throughput process development and other combinatorial approaches are discussed and compared according to their experimental effort and understanding. With a growing sea of information, novel modelling tools and artificial intelligence (AI) gain more importance to find patterns behind the data and thereby acquire a deeper process understanding.

Keywords: downstream processing, vaccine purification processes, model-based high throughput process development, artificial intelligence

Glossary

[table 3]

1. Introduction

The COVID-19 pandemic is engulfing the world, which already costs over millions of lives and is still infecting hundreds of thousands of people daily one and a half year after the first outbreak in December 2019. More than ever the world is aware of the value of vaccination contributing to improved public health, reduced healthcare costs, economic growth, travel safety and prolonged life expectancy (Andre et al., 2008; Rodrigues & Plotkin, 2020). In general, vaccination is estimated to prevent 2-3 million childhood and almost 6 million adult deaths annually (Andre et al., 2008; Ehreth, 2003). Recently, the WHO published an action plan making vaccination available to everyone in the world and promoting innovation within the vaccine industry (Arora et al., 2014).

High safety and purity demands increase of the complexity of the vaccine purification process and as a consequence making the downstream process very expensive in both costs and time (Wen et al., 2014; Zhao et al., 2019). The downstream process plays a key role in reducing contaminants to very low values in order to prevent for example high reactogenicity or unwanted immune responses and guarantee the safety and efficacy of the vaccine. Designing a vaccine purification process is accompanied with many decisions, such as type and sequential order of purification techniques, conditions, costs and other performance measurements (Pirrung et al., 2019). Additionally, optimization of a single unit operation and overall purification sequence is important, whereas small variations, regarding conditions, in certain steps may affect the subsequent unit operation performance. Fast vaccine process development is of utmost importance in light of infectious outbreaks and pushing competitive market and it highly depends on its design strategy for a purification process (Wen et al., 2014). Traditionally, vaccines are developed within 10 – 15 years, hence pharmaceutical companies desire to reduce the process development time drastically in every aspect. One of the reasons why the first SARS-CoV-2 vaccines could be developed within 1-2 years, is the employment of an accelerated development timeline due to parallelization of phases instead of sequential development (Krammer, 2020). Additional arguments for such a quick development are knowledge from previous research on related viruses as well as adapted production processes from existing vaccines (i.e. platform knowledge), massive funding to enable parallelization, risk-taking and fast regulatory reviewing (Ball, 2021).

The ‘quality by design’ (QbD) paradigm (FDA, 2004; ICH, 2009) is one of the main reasons that made the pharmaceutical industry shift from a trial-and-error approach towards a more comprehensive systematic and efficient approach with the purpose to increase process understanding and process control (Hanke & Ottens, 2014; Lacki, 2014; Rathore, 2009, 2016; Yu, 2008). The implementation of high throughput process

development (HTPD) approaches contributes to faster and more efficient process developments while using less material and hence being more cost-effective (Hanke & Ottens, 2014). HTPD is a combinatorial approach of both high throughput experimentation (HTE) and often modeling techniques. Recently Sao Pedro et al. (São Pedro et al., 2021) outlined the major problems (e.g. cell culture, filtration and analytical tools) within HTPD along with suggested solutions (microfluidics, modeling and Process Analytical Technologies (PAT)) for the purpose of integrated and continuous bio manufacturing. Although this review is not focused on continuous biomanufacturing, the current limitations of HTPD are likewise applicable to the vaccine purification process development.

Vaccine purification processes can differ enormously from each other as they highly depend on the type of vaccine and crude starting material/host organism (e.g. fertilized eggs, bacterial-, mammalian-, and insect cells). Carvalho et al. (Carvalho et al., 2021) pointed out the influence of vaccine types on downstream process strategies and described into detail each vaccine purification step with a focus on influenza vaccines. A general overview of vaccine types is shown in Figure 1 being classified as whole pathogen (inactivated or attenuated), antigenic components (subunit) of pathogen or nucleic acid vaccines, though slightly different classifications have also been reported.

[Figure 1]

While the downstream process is minimal for live and inactivated vaccines in order to preserve their genetic stability, the purification of recombinant or subunit vaccines requires a complex purification challenge because of closely related host cell proteins HCP removal with respect to the target protein (Wen et al., 2014). Recently Jones et al. (Jones et al., 2021) pointed out the concerns of high-risk HCPs and recommended a strategy for monitoring and eliminating the known impurities. Despite the great variance between different vaccine downstream processes, the generic order of purification steps is similar as shown in Figure 2, note that cell lysis is only required if the antigen produced is intracellular. Detailed purification schemes for certain vaccine types are outside the scope of this paper and can be found elsewhere. Josefsberg and Buckland (Josefsberg & Buckland, 2012) described the production process of several virus-based conjugate and DNA vaccines, while Abdulrahman and Ghanem (Abdulrahman & Ghanem, 2018) summarized the most recent advances in the purification of plasmid DNA vaccines. In the book of Wen et al. (Wen et al., 2014), viral vaccines purification (Kalbfuss-Zimmermann & Reichl, 2014) and protein subunit vaccines purification (Yang & D'Amore, 2014) are described into more detail.

[Figure 2]

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3 107 Most of the current vaccine development approaches are based on design of experiments (DoE), in which
4 108 multiple factors are changed simultaneously to evaluate the underlying interactions and hence obtaining a
5 109 multidimensional model that correlates the effects of various factors on the critical quality attributes (CQA),
6 110 which is an essential aspect within QbD guidelines (Kumar et al., 2014; Rathore, 2016). However, the exist-
7 111 ing vaccine process development strategy requires high experimental effort and little process understand-
8 112 ing is gained. Moreover, the sequential determination of purification steps and individual process optimi-
9 113 zations might lead to a suboptimal process design with respect to the objective such as yield or costs (Huuk
10 114 et al., 2014; Otero et al., 2014; Pirrung et al., 2017). A standardized approach, also known as platform
11 115 process, as established for monoclonal antibodies (mAbs) (Shukla & Thommes, 2010), is yet missing mainly
12 116 due to the large diversity between vaccine types and their complexity. A platform process for specific vac-
13 117 cine types would be highly beneficial in terms of process development time, knowledge, resources, costs
14 118 and regulatory aspect (Zhao et al., 2019). Another often complicated task is the precision of quantitatively
15 119 measuring and characterizing virus or bacterial particles and in addition the lack of rapid analytical technol-
16 120 ogies (Kalbfuss-Zimmermann & Reichl, 2014; Zhao et al., 2019). A trend within the QbD initiative is the use
17 121 of PAT allowing real-time measurements to ensure consistent product quality and performance, besides
18 122 providing a better understanding of the process (Rathore, 2016). In addition, mechanistic models rely on
19 123 physical processes occurring during a certain separation step and can therefore be of great merit to in-
20 124 crease process understanding, but also decrease experimental effort by simulating experiments and allow-
21 125 ing to perform processes on different scales *in-silico*. The use of AI techniques could eliminate shortcomings
22 126 within the modeling area and bring modeling techniques to a higher level of applicability and usability.
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37 127 This review presents modern and future downstream process development approaches and their applica-
38 128 tion in (bio)pharmaceutical industry with a focus on chromatography as this is the main purification tech-
39 129 nique for (subunit) vaccines. This paper aims to show the evolvement of model-based high throughput
40 130 process development approaches by making use of more advanced modeling techniques, such as empirical,
41 131 mechanistic and hybrid modeling. The applicability and benefit using these methods are supported by case
42 132 studies from industry and academia.
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2. Downstream process development methods

The overall goal of process development is to design the optimal purification process, by means of achieving purity targets, at minimum costs and time efforts, by at the same time adhering to all regulatory requirements. Currently, vaccine development employs mostly DoE-based methods, though it could benefit from more advanced model-based process development approaches, which are already used in other biopharmaceutical branches, such as for the purification of mAbs. **Figure 3** shows two types of process development approaches, the DoE-based method and a modeling-based method. In the following section, process development approaches are described briefly. More comprehensive reviews on this topic can be found elsewhere (Baumann & Hubbuch, 2017; Hanke & Ottens, 2014).

[Figure 3]

2.1. Experimental driven downstream process development

2.1.1. One-factor-at-a-time and Design of Experiments

One-factor-at-a-time (OFAT) is a more traditional approach in which one factor is changed during a series of experiments while the other factors are kept constant. In this method dependencies between factors are neglected and therefore discovering the optimum is rather difficult and quite inefficient (Czitrom, 1999). For that reason, the biopharmaceutical industry shifted a while ago to design of experiments (DoE) based upon statistical methods to design and analyze experiments and so to obtain more valuable information by conducting less experiments. The classical DoE-method is factorial design, performing experiments on all possible combinations at every factor level with the purpose to identify effects of each factor as well as interactions between factors on the response. An improvement on the classical DoE-screenings is definitive screening design which estimates the curvature effects and enables to separate factors having a significant impact on the response from the factors having negligible effects. Other methods offering a three level multifactorial design are for example Box-Behnken (Box & Behnken, 1960) or central composite designs. Hibbert D.B. (2012) extensively described the most common used DoE methods with a focus on chromatography (Ferreira et al., 2007; Hibbert, 2012). Various DoE software are available nowadays, such as Design-Expert, Modde and JMP, though other statistical software, like R and SPSS, can also be used for DoE purposes.

2.1.2. Parameter acquisition for modelling purposes

An alternative experimental strategy is to determine parameters that serve as input for mechanistic or physical models. The use of mechanistic models has been established decades ago and nowadays adopted widely by chemical industry who are even designing their processes entirely *in-silico* (Siirola, 1996). Only recently, biopharmaceutical industry initiated this strategy into their process development, in which the major challenge, especially for vaccines, is often the complexity of feed mixture, which contains the product of interest (e.g. antigen) together with thousands of proteins and impurities (Yang & D'Amore, 2014). However, HTE made it worthwhile to determine model parameters even for more complex mixtures (Benedini et al., 2020; Hanke et al., 2016; Pirrung et al., 2018). Noteworthy, a validated model increases process understanding and enables to optimize the process *in-silico*, resulting in time, material and costs savings (Nfor et al., 2009). For chromatographic purposes, as this is the main purification technique in vaccines production, the adsorption isotherm parameters, describing the binding behavior of components to the solid phase, are of utmost importance. Experimental determination of adsorption equilibria is required to regress the isotherm parameters and can be obtained by batch adsorption experiments (Carta & Jungbauer, 2010; Chen & Cramer, 2007; Moreno-González et al., 2020; Nfor et al., 2010; Pirrung et al., 2018), frontal analysis, isocratic, - or linear gradient elution (Brooks & Cramer, 1992; Carta & Jungbauer, 2010; Osberghaus et al., 2012) or by making use of inverse techniques, which optimizes the difference between experimental and simulated elution profiles by tuning certain parameters (Osberghaus et al., 2012; Pirrung et al., 2018; Saleh et al., 2020). Besides isotherm determination, column and resin characteristics have also to be obtained in order to acquire a validated model, however these are not as difficult to obtain.

2.1.3. High Throughput Screening (HTS)

The introduction of liquid handling stations (LHS), about two decades ago, allowed the acceleration of conducting experiments, also known as HTE or HTS. Due to automation, miniaturization and parallelization it was viable to create large number of data sets while using a reduced amount of sample volume and resources within a shorter time-frame (Bhambure et al., 2011; Wiendahl et al., 2008). Another benefit of automation is the lowered variability and superior precision (Singh & Herzer, 2018). Nowadays, LHS is a widely applied technique in both academia and industry and reduced significantly the process development time and feasibility to reach an optimal overall process performance (Bensch et al., 2005; Bhambure & Rathore, 2013; Rathore et al., 2018). Apart from these system's benefits there are certainly also some disadvantages as pointed out elsewhere (Bergander & Lacki, 2016; Lacki, 2012). For example its limitation in

accurately mimicking the flow distributions of process columns (Rathore et al., 2018). HTS requires high understanding of designing your experiments efficiently in order to make optimal use of the system, therefore it is rather a tool to be used than an approach on its own.

2.2. Expert-knowledge driven downstream process development

2.2.1. Universal

Rules of thumb, available knowledge and experience of previous processes are the basis for expert knowledge or heuristic approaches to design new product processes (Baumann & Hubbuch, 2017; Lienqueo & Asenjo, 2000). Using experts insights is easy to apply and could speed up the process design by eliminating combinations of unit operations with less promising results (Nfor et al., 2008). Asenjo et al. (Leser & Asenjo, 1992; Lienqueo & Asenjo, 2000) developed an expert system focused on downstream protein processes; this software uses databases consisting of expert knowledge on universal process designs (heuristics) to support and accelerate decision making for the selection of a sequence of unit operations. Several handbooks, like Sofer & Hagel (Hagel et al., 2008) and GE healthcare (Sciences, 2012), outline general design heuristics extensively. Most vaccine purification processes are also based upon heuristics, as for example the purification of hepatitis A virus from mammalian cell cultures, in which the first step involves a low-cost anion-exchange chromatography to capture the product and remove a substantial amount of impurities and the last step of the downstream process a polishing and desalting step using size-exclusion chromatography (Hagel et al., 2008; Hagen et al., 2000; Kalbfuss-Zimmermann & Reichl, 2014). A general example that is almost entirely based on knowledge are platform processes as explained into more detail in the next paragraph.

2.2.2. Platform process

Platform processes are used as 'templates' for designing an entire purification sequence for a specific type of molecule, utilizing a pre-established series of unit operations (Baumann & Hubbuch, 2017). The platform instructions provide, corresponding to the overall purification process, details of the operating conditions for each unit operation. One of the key advantages is a reduced process development time, regulatory aspect and resources for similar molecules and accordingly decreasing time-to-market and validation effort (Hagel et al., 2008). Moreover, the platform documents can be shared and aligned among different departments but also across different manufacturing sites, serving as a site-independent process (Shukla et al.,

2007). However, biopharmaceuticals with similar characteristics and thus purification steps are more suited for a platform process approach (Hagel et al., 2008; Shukla & Thommes, 2010). For example, mAbs are relatively well defined and platform processes are used to establish similar purification processes for new mAbs variants. Detailed information about process-related contaminants such as persistent HCPs and other impurities for the corresponding cell culture, i.e. CHO and hybridism are known (Shukla et al., 2007). The order of purification steps includes protein A chromatography, low pH viral inactivation, IEX chromatography polishing steps, viral filtration and ultra-, diafiltration. Only small changes are required in the purification process conditions to determine a ‘new’ mAb variant purification process. Other potential candidates for platform approaches could be pDNA vaccines and influenza vaccines, both having similar properties and purification steps (Abdulrahman & Ghanem, 2018; Hagel et al., 2008). However, mAbs are relatively similar to each other in their properties, while vaccines vary greatly in their appearance, making it more difficult to standardize the purification process.

2.3. Model-based downstream process development

In process engineering models play an important role, they aim to represent a real system in an abstracted mathematical format (Muller et al., 2012; von Stosch et al., 2014). Bézivin and Gerbé (Bezivin & Gerbe, 2001) defined a model as “a simplification of a system built with an intended goal in mind. The model should be able to answer questions in place of the actual system”. The intended goal related to process engineering could be for example, control, simulation, design, monitoring or optimization, and depending on the goal a different model is appropriate (Bonvin et al., 2016). Models help to understand complex problems and could provide potential solutions if the model is an adequate representation of the modelled system’s features of interest (Selic, 2003). Running the model with a given set of parameters is a simulation and hence an inexpensive and safe way to run a virtual experiment (Ljung & Glad, 1994). For that reason, the number of experiments in laboratory can be reduced and/or designed more efficiently, thereby reducing time and material consumption. In order to build a model two main sources are essential, the knowledge translated into laws of nature and the collection of data obtained from the real system (Ljung & Glad, 1994). In process engineering, a distinction can be made between first-principles, mechanistic or knowledge-driven models and data-driven or empirical models respectively known as transparent white-box and less transparent black-box models (von Stosch et al., 2014). A combination of both is named hybrid semi-parametric models. An overview of the main advantages and disadvantages is given in Table 1.

[Table 1]

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256 2.3.1. Data-driven models

257 Data-driven or empirical models attempt to describe the input-output relation based upon observed ex-
258 periments within a predefined design space, such as artificial neural networks (ANN), statistical and regres-
259 sion models (Bonvin et al., 2016). The biopharmaceutical industry makes often use of statistical models,
260 either by executing a predefined set of experiments using DoE and an appropriate statistical method, such
261 as response surface methodology (RSM), to evaluate the data or by employing a multivariate data analysis
262 using an existing dataset (Rathore et al., 2014). RSM is a well-known empirical model and describes the
263 relation of a response between different tested factors within a DoE and a model attempts to describe the
264 mathematical relationship (Hibbert, 2012). This statistical (black-box) model solely observes the factor to
265 response correlation without gaining fundamental mechanistic (physiochemical) understanding of the es-
266 timated parameters. By making use of DoE and regression analysis through first- and second order polyno-
267 mials the optimum input combination can be estimated (Kleijnen, 2015). However, fitting data to second
268 order polynomials is a major drawback of RSM as frequently not all curvatures within the systems can be
269 described by the second order polynomial (Baş & Boyacı, 2007). DoE in combination with empirical model-
270 ling has been widely applied to design downstream purification processes in biopharmaceutical industry
271 and academia (Anirban Roy et al., 2013; Eon-Duval et al., 2003; Touelle et al., 2011). The effect of high-salt
272 solution on RNA precipitation and pDNA recovery was investigated using DoE and linear regression models
273 (Eon-Duval et al., 2003). And more recently, Chiang et al. (Chiang et al., 2019) evaluated the impact of
274 chromatographic parameters on virus clearance when switching from a single to multicolumn operation
275 utilizing DoE. A major limitation of data-driven models is that they are merely valid in a defined region of
276 measured variables and only able to predict variables in that region, making extrapolation generally highly
277 inaccurate. Moreover, little process knowledge can be extracted, because the parameters are often just
278 correlations (Solle et al., 2017). On the other hand, data-driven modeling requires no process understand-
279 ing in advance and is less time consuming compared to mechanistic modeling (Solle et al., 2017).

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2.3.2. Mechanistic models

Mechanistic, first-principle, or knowledge-driven models attempt to describe the inner mechanisms and phenomena occurring in a process or system based upon knowledge about the process. These models consist of material and/or energy balances together with transport and thermodynamic phenomena and have a fixed structure, meaning the parameters might have a physical interpretation (Solle et al., 2017). The model parameters are estimated by experimental data or physical correlations. The physical processes occurring during a purification process can be translated into mathematical simulation models. A validated mechanistic model allows to explore various conditions *in-silico* and therefore enables to acquire optimum operating conditions more efficiently (Pirrung & Ottens, 2017). The phenomena taking place inside a chromatographic column are well described in literature, Ruthven (Ruthven, 1984) extensively outlined the dynamics and adsorption processes. Kinetic or rate models are most common in practice, including dispersive factors, like mass transfer and dispersion effects, and equilibrium factors, such as adsorption isotherms, ionic dissociation and intermolecular association (Carta & Jungbauer, 2010). The three most prominent kinetic models, general rate-, lumped pore-, and lumped kinetic model, vary in the degree of covering the pore diffusion effect in decreasing order and corresponding complexity (Felsing & Guiochon, 2004). However, for all mechanistic models applies that isotherm parameters are crucial as explained previously in section 2.1.2, for which numerous binding models exist, such as linear, Langmuir (Langmuir, 1916), steric mass action (Brooks & Cramer, 1992) and mixed-mode (Nfor et al., 2010). The utilization of chromatographic models varies from process synthesis, optimization and control (Andersson et al., 2017; Gomis-Fons et al., 2020; Moreno-González et al., 2021; Nfor et al., 2012b; Papathanasiou et al., 2017; Shekhawat et al., 2017; Westerberg et al., 2012) to scale-up, resin selection and robustness checks (Close et al., 2014; Liu et al., 2017; Nfor et al., 2011b; Vogg et al., 2020). One step further is to simulate a combination of integrated chromatography and other conditioning steps to find the overall optimum purification process (Hamidi et al., 2016; Huuk et al., 2014; Löfgren et al., 2019; Nfor et al., 2013; Pirrung et al., 2019; Schmölder & Kaspereit, 2020). Nowadays, various commercial software of chromatographic mechanistic models are available, e.g. ChromX (Hahn et al., 2015), Aspen Chromatography , DelftChrom , CADET (Leweke & von Lieres, 2018) and ChromaTech (Meyer et al., 2020).

Alternative *in-silico* methods for adsorption experiments have been investigated for several years. Molecular dynamics simulations attempt to describe the interaction between resin-proteins on a detailed atomic level (Banerjee et al., 2017; Dismer & Hubbuch, 2010; Parimal et al., 2015). Quantitative structure-activity

relationships (QSAR) combines molecular properties with empirical modeling to find correlations amongst retention behavior and protein surface properties (Kittelmann et al., 2017a, 2017b; Woo et al., 2015). This kind of molecular modeling can be used to predict the retention behavior of proteins on resins to reduce process development times (Buyel et al., 2013). However, often detailed information is required about each component, such as amino acid sequence or crystal structure and also a large amount of experiments (Pirrung & Ottens, 2017).

Mechanistic models enable to explore conditions over a wide range and even beyond the observed measured results, accordingly showing an increased extrapolation capability compared to data-driven models (Solle et al., 2017). This contributes to process understanding, which is line with the QbD initiative, although mechanistic modeling also requires process understanding. The major drawback of knowledge-based models is complexity and hence requires more development time compared to data-driven models.

2.3.3. Hybrid (semi-parametric) models

Hybrid (semi-parametric) modelling combines parametric (i.e. first principle-, mechanistic-, and knowledge-based models) with nonparametric (i.e. data-driven models) in order to eliminate drawbacks of individual approaches and get the best out of both (von Stosch et al., 2014). Von Stosch et al. extensively reviewed the hybrid semi-parametric modelling framework and the various applications in (bio)chemical engineering concerning process monitoring, control, optimization, model-reduction and scale-up. The configuration when using both parametric and nonparametric models can be serial or parallel depending on the scope of the model. Usually a parallel mode is recommended when the mechanistic (white-box) model performance is limited or insufficiently accurate and the addition of a nonparametric (black-box) model may improve the estimations, [Figure 5](#). A serial approach is often utilized for reducing complexity of mechanistic (white-box) models by determining parameters using nonparametric (black-box) models or the results of mechanistic (white-box) models can function as an input for nonparametric (black-box) models (von Stosch et al., 2014).

[\[Figure 5\]](#)

The utility of hybrid modeling is the cost-effective benefit in terms of solving a complex problem and to develop a model more efficiently. Other advantages are an improved model accuracy, transparency and extrapolation properties, besides gaining a broader process understanding (Solle et al., 2017).

3. High Throughput Process Development

3.1. Single or double purification steps

Hybrid process development approaches combine experimental and modelling tools to design a process. After the introduction of LHS, hybrid approaches gained a special interest as LHS enabled to perform experiments in high throughput manner. Utilizing HTE in relation to process design is known as HTPD, combining HTS with empirical or mechanistic modeling is named model-based HTPD (Hanke & Ottens, 2014; Nfor et al., 2009). The implementation of HTPD pursues the QbD paradigm in terms of process and product understanding and hence contributing to high and stable product quality as well as process robustness (Bhambure et al., 2011). The establishment of HTPD arose about 15 years ago (Bensch et al., 2005; Nfor et al., 2008; Rege et al., 2006; Susanto et al., 2008) and evolved ever since as an efficient and cost-effective-ness method, broadly acknowledgement by industry (Lacki, 2014). (Model-based) HTPD can be applied in various development stages and for different purposes, like resin and solubility screenings, design-space definition, risk-assessment, process robustness and control. In the review of Baumann and Hubbuch (Baumann & Hubbuch, 2017) several commercial miniaturized HT-suitable systems in both up,- and down-stream process development are described. The technical review of Lacki (Lacki, 2012) outlined the most frequently used chromatography HT equipment, such as microtiter filter plates, prefilled pipette tips and robocolumns, nowadays ranging from 50-600 μ L. Bhambure and Rathore (Bhambure & Rathore, 2013) proved a tremendous increase in productivity (170 x higher) utilizing a HTPD platform (2 and 6 μ L resin volume) against the traditional laboratory scale (0.5 mL resin volume) for defining the characterization space of an ion exchange chromatography step using DoE. A more practical and general HTPD workflow was developed by Welsh et al. (Welsh et al., 2014) involving a multistep approach of HT chromatography techniques as a guidance for defining the operating space. No detailed modeling tools were implemented as accurate performance predictions were not the aim, only isotherm models to regress the partitioning coefficient and maximum binding capacity were used. Weigel et al. (Weigel et al., 2019) applied a similar method as Welsh et al. to investigate the effectiveness of hydrophobic interaction chromatography (HIC) as a final purification step for a cell culture-derived influenza A and B virus. 96-well filter plate experiments were used for screening various resins and salt concentrations, followed by conventional lab-scale columns for dynamic binding capacity characterization. However, the major reason for choosing a rational step-wise method over mathematical modeling was the lack of available virus purification data by HIC to be able to determine model parameters. As vaccine platform processes are barely available yet, Ladd Effio et al. (Ladd Effio et al., 2016) initiated a capture step as first part of a generic purification platform process for virus-

like particles (VLP). Ladd Effio et al. (Ladd Effio et al., 2016) established a one-step removal of HCPs and DNA from a complex VLP feedstock with an anion-exchange membrane capture step by making use of HTE and mechanistic modeling for *in-silico* optimization purposes. Although equilibrium and binding capacities of membrane chromatography are often limited, at high flowrates membrane chromatography outperforms conventional packed bed chromatography in terms of productivity and for short residence times also in bed utilization (Boi et al., 2020). It is expected that in the near-future membrane materials with higher binding capacities will become available and therefore could overcome the restriction on surface area per unit volume of resin. The advancement in membrane chromatography technology is definitely interesting to the biopharmaceutical industry.

Even though chromatography is one of the main purification techniques for biopharmaceuticals and vaccines, other downstream process techniques are also HT suited. Precipitation is a well-known technique to isolate a desired component such as a protein, DNA or virus and proven to be HT suited (Nfor et al., 2011a) (Kalbfuss-Zimmermann & Reichl, 2014). This separation technique depends on the physical and/or chemical interaction between the precipitating agent (e.g. calcium chloride, ammonium sulfate or PEG) with one or several of the components in which solubility is the most critical thermodynamic property (Lovrien & Matulis, 1997). Aqueous two-phase systems (ATPs) could also be an alternative to chromatography as it is based on liquid-liquid extraction employing two immiscible phases to separate components from mixtures. HT techniques in combination with statistical (Oelmeier et al., 2011; Zimmermann et al., 2016) or mathematical/thermodynamic (Bussamra et al., 2021) models are a convenient method for characterizing these systems.

Analytics to monitor the process are just as important as the purification techniques itself. Analytics however remains a bottleneck during HTE, and consequently slows down experimentation considerably. Konstantinidis et al. (Konstantinidis et al., 2012) provided a strategic assay deployment that helps selecting appropriate analytical methods, while preserving data quality. Nonetheless, finding innovative ways to accelerate the analytical throughput would be of great merit.

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403 [Table 2]

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3.2. Overall purification process

The studies described in the previous paragraph focused mainly on applying HTPD to one or two sequential purification steps, but thereby does not consider the overall purification workflow. Designing a downstream process by optimizing each unit operation individually could lead to a suboptimal process design as small variations in one-unit operation may affect the performance of subsequent following purification steps. The combination of HT and model-based optimization approaches for a sequence of unit operations has seldom been studied. Nfor et al. (Nfor et al., 2013) established a systematic approach to rationally define the protein purification process utilizing a top-to-bottom approach. The least promising flowsheets were eliminated at each tree-diagram level by means of flow-sheet selection with the aim of keeping a minimum number of purification units. Instead of sequential optimization, which might generate a suboptimal process (Huuk et al., 2014; Otero et al., 2014), Huuk et al. (Huuk et al., 2014) presented a simultaneous two-step ion exchange chromatography process flowsheet optimization, including salt-gradient shapes and cut-points for fraction collection. Nonetheless, Pirrung et al. (Pirrung et al., 2019) even proved the feasibility to simultaneously optimize an integrated process considering of three chromatographic steps (e.g. cation exchange, hydrophobic interaction and mixed-mode) including buffer exchange steps in between (e.g. ultra- and diafiltration) applied to a complex biological feedstock purification. First the isotherm parameters were acquired utilizing HT techniques as illustrated in more detail in previous work (Pirrung et al., 2018), hence other parameters were obtained by conventional lab-scale experiments. The use of ANN to find sufficient starting conditions for the local optimization using mechanistic models enabled to circumvent speed-limitations (Pirrung et al., 2019; Pirrung et al., 2017).

4. Artificial Intelligence in Process Development

The interest in HTPD raised after the introduction of HT technologies, having the major benefit to generate more data while consuming less material. Nevertheless, these arising technologies still face a number of hurdles. Experimentally transferring every item into HT mode including analytics remains a burden and although more data is being produced, processing and handling these data efficiently is still challenging. Modeling is a promising tool to close this gap. Further advancements of modeling are discussed in the following paragraph.

While complex mechanistic models attempt to describe the mechanisms and thermodynamic phenomena, determining certain parameters is rather difficult. Simplifying models could avoid certain difficulties however oversimplifications may cause inaccurate predictions and meaningless results. The optimal model should be as simple as possible while still gaining high or sufficient understanding. Moreover, a trade-off between accuracy versus speed has to be made especially when running optimizations. This led to the question how to reduce the computational time effort or simplify complex models while retaining a similar level of accuracy and/or understanding.

Due to the generation of larger data-sets and better computer systems in recent years, the use of AI gained popularity in various technology fields, likewise within the biotechnology area. Albeit, it was already in the late 90's that ANN were used to predict retention times in chemical chromatography (Havel et al., 1999; Marengo et al., 1998). Moreover, in 1992 Psychogios and Ungar (Psychogios & Ungar, 1992) presented the first hybrid neural network-first principles approach applied to model a fed-batch bioreactor. This hybrid model used a neural network model to estimate unknown process parameters serving as an input to a first principle model, resulting in an improved inter- and extrapolating capability and understanding over merely "black-box" neural networks. Von Stosch et al. (von Stosch et al., 2014) extensively reviewed the hybrid semi-parametric modelling framework, as explained in 2.3.3., and the various applications in (bio)chemical engineering concerning process monitoring, control, optimization, model-reduction and scale-up. Nagrath et al. (Nagrath et al., 2004) established an optimization framework using a serial white- and black-box configuration (Figure 5) to find the optimal design for a chromatographic process applied to a binary and tertiary mixture. After obtaining experimentally the physical model parameters, numerous simulations were performed under various conditions using the physical model (i.e. white-box) for training the neural network. Finally, the optimal operating conditions for several purity levels were identified by using the neural network to advance the computation. Likewise, Pirrung et al. (Pirrung et al., 2019; Pirrung et al., 2017) used ANN to accelerate a flowsheet optimization consisting of three chromatography and UF/DF units. However,

here ANN (black-box), along with a global optimization algorithm, was used to find adequate starting conditions for the local optimization that was performed together with a mechanistic model (white-box) in order to assure realistic and accurate results. A speed improvement of 70% was found, including training of the networks, compared to using solely mechanistic models for the optimization. Recently, Nikita et al. (Nikita et al., 2020) showed a novel approach making use of reinforcement learning (RL) to increase computational efficiency during a continuous chromatography process optimization. Each mechanistic model simulation (white-box) is rewarded according to a RL-method and consequently the optimization criteria (design space) are adjusted to accelerate the convergence of optimization. The optimal flowrate, directly related to yield and purity demands, was found three times faster using the RL based optimization method compared to conventional trial and error methods. Apart from using hybrid semi-parametric modelling for optimization intentions, other research showed the usefulness of black-box modelling to estimate certain white-box model parameters that are hard to determine. Wang et al. (Wang et al., 2017) used neural networks to directly derive mass transfer, isotherm and characteristic charge parameters from experimental chromatograms, thereupon these parameters served as input for the mechanistic model (white-box). In this way time-consuming experimental methods to determine these parameters were circumvented. In mechanistic filtration models the flux is a key parameter, however predicting this parameter accurately might be quite complex. Therefore Krippel et al. (Krippel et al., 2020) used an ANN (black-box) to determine the flux using transmembrane pressure, cross-flow velocity and concentration as input parameters. Placing the hybrid model in series enabled to perform a multistep ahead prediction of the concentration over time. In general, data-driven models in addition to white-box models or vice versa can be advantageous in terms of prediction accuracy, computing and model development efficiency and enhanced extrapolation properties (von Stosch et al., 2014).

With an eye on the future more applications of hybrid modelling approaches in both industry and academia are expected instead of solely employing one modelling approach. Moreover, the modelling techniques utilized in the process development (HTPD) can also be used for process control and optimization in later development stages and manufacturing processes. One step ahead is industry 4.0 known as the latest revolution and aiming to digitalize the whole manufacturing process, from process control to decision-making and efficiently collecting all monitored data that in turn is also valuable for process development (Markarian, 2018). In order to realize Industry 4.0, digital twins are highly essential, defined as a virtual counterpart of the physical process and their interconnection (Portela et al., 2020).

5. Summary and Conclusion

Vaccination protects millions of people against infectious diseases and because a high product quality is pivotal, the downstream processing is likewise as important. Downstream process operations in manufacturing have a direct influence on time-to-market, product quality and cost of goods and hence modernizing the strategies for developing processes could be of great merit. The urge to decrease the process development timeline of vaccines raised as well as the need for deeper process understanding as stated by the quality by design guideline.

The introduction of high throughput technology accelerated experimental data generation and allowed to investigate the influence of parameters more thoroughly and systematically. However, HT also required to enhance data-processing and modeling techniques. Mechanistic models provide insights on the inner working mechanism of unit operations and are being adopted more by industry in recent years, proving they add deeper process understanding and greater application possibilities. The combination of HT and modeling techniques led to HTPD approaches, acquiring and using data in a more efficient and purposeful way and thereby also enabling standardized process development approaches. The future direction in process development is to design and optimize the overall downstream process *in-silico* for which only a limited number of model calibration and validation experiments are needed. Hybrid (semi-parametric) modeling can help to ease the model development or improve the accuracy by making optimal use of both mechanistic and data-driven models. Recent research has shown the potential of artificial neural networks in addition to mechanistic models or vice versa to circumvent speed-limitation or estimation of parameters.

With these emerging new technologies, it will now be possible to standardize process development workflows provided that a proficient combination of experimenting and modeling techniques is utilized. Creating a generic process development workflow will enhance process development time and shared knowledge among different departments and manufacturing sites.

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517 Conflict of interest

518 All authors have declared the following interests: Geoffroy Geldhof and Olivier Le Bussy are employees of
519 the GSK group of companies. The other authors declare no conflict of interests.

520

521 Author contributions

522 Daphne Keulen and Marcel Ottens were involved in the conceptualization of the manuscript, Daphne Keu-
523 len wrote the original draft, all authors contributed to the development of the manuscript, reviewed and
524 edited the final manuscript.

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6. References

- Abdulrahman, A., & Ghanem, A. (2018). Recent advances in chromatographic purification of plasmid DNA for gene therapy and DNA vaccines: A review. *Analytica Chimica Acta*, 1025, 41-57. doi:10.1016/j.aca.2018.04.001
- Andersson, N., Löfgren, A., Olofsson, M., Sellberg, A., Nilsson, B., & Tiainen, P. (2017). Design and control of integrated chromatography column sequences. *Biotechnology Progress*, 33(4), 923-930. doi:<https://doi.org/10.1002/btpr.2434>
- Andre, F. E., Booy, R., Bock, H. L., Clemens, J., Datta, S. K., John, T. J., Lee, B. W., Lolekha, S., Peltola, H., Ruff, T. A., Santosham, M., & Schmitt, H. J. (2008). Vaccination greatly reduces disease, disability, death and inequity worldwide. *Bulletin of the World Health Organization*, 86(2), 140-146. doi:10.2471/BLT.07.040089
- Anirban Roy, C., Paramita, B., & Gandham, S. P. (2013). Development of suitable solvent system for downstream processing of biopolymer pullulan using response surface methodology. *Plos One*, 8(10). doi:10.1371/journal.pone.0077071
- Arora, N., Al Mazrou, Y., Cravioto, A., & al., e. (2014). 2014 Assessment report of the global vaccine action plan.
- Ball, P. (2021). The lightning-fast quest for COVID vaccines - and what it means for other diseases. *Nature*, 589, 16- 18. doi:<https://doi-org.tudelft.idm.oclc.org/10.1038/d41586-020-03626-1>
- Banerjee, S., Parimal, S., & Cramer, S. M. (2017). A molecular modeling based method to predict elution behavior and binding patches of proteins in multimodal chromatography. *Journal of Chromatography A*, 1511, 45-58. doi:<https://doi.org/10.1016/j.chroma.2017.06.059>
- Baş, D., & Boyacı, İ. H. (2007). Modeling and optimization I: Usability of response surface methodology. *Journal of Food Engineering*, 78(3), 836-845. doi:<https://doi.org/10.1016/j.jfoodeng.2005.11.024>
- Baumann, P., & Hubbuch, J. (2017). Downstream process development strategies for effective bioprocesses: Trends, progress, and combinatorial approaches. *Engineering in Life Sciences*, 17(11), 1142-1158. doi:10.1002/elsc.201600033
- Benedini, L. J., Figueiredo, D., Cabrera-Crespo, J., Gonçalves, V. M., Silva, G. G., Campani, G., Zangirolami, T. C., & Furlan, F. F. (2020). Modeling and simulation of anion exchange chromatography for purification of proteins in complex mixtures. *Journal of Chromatography A*, 1613, 460685. doi:<https://doi.org/10.1016/j.chroma.2019.460685>
- Bensch, M., Schulze Wierling, P., von Lieres, E., & Hubbuch, J. (2005). High Throughput Screening of Chromatographic Phases for Rapid Process Development. *Chemical Engineering & Technology*, 28(11), 1274-1284. doi:<https://doi.org/10.1002/ceat.200500153>
- Bergander, T., & Lacki, K. M. (2016). High-throughput process development: Chromatography media volume definition. *Engineering in Life Sciences*, 16(2), 185-189. doi:<https://doi.org/10.1002/elsc.201400240>
- Bezivin, J., & Gerbe, O. (2001, 26-29 Nov. 2001). Towards a precise definition of the OMG/MDA framework. Paper presented at the Proceedings 16th Annual International Conference on Automated Software Engineering (ASE 2001).
- Bhambure, R., Kumar, K., & Rathore, A. S. (2011). High-throughput process development for biopharmaceutical drug substances. *Trends in Biotechnology*, 29(3), 127-135. doi:10.1016/j.tibtech.2010.12.001
- Bhambure, R., & Rathore, A. S. (2013). Chromatography process development in the quality by design paradigm I: Establishing a high-throughput process development platform as a tool for estimating

- “characterization space” for an ion exchange chromatography step. *Biotechnology Progress*, 29(2), 403-414. doi:<https://doi.org/10.1002/btpr.1705>
- Boi, C., Malavasi, A., Carbonell, R. G., & Gilleskie, G. (2020). A direct comparison between membrane adsorber and packed column chromatography performance. *Journal of Chromatography A*, 1612, 460629. doi:<https://doi.org/10.1016/j.chroma.2019.460629>
- Bonvin, D., Georgakis, C., Pantelides, C. C., Barolo, M., Grover, M. A., Rodrigues, D., Schneider, R., & Dochain, D. (2016). Linking Models and Experiments. *Industrial & Engineering Chemistry Research*, 55(25), 6891-6903. doi:10.1021/acs.iecr.5b04801
- Box, G. E. P., & Behnken, D. W. (1960). Simplex-Sum Designs: A Class of Second Order Rotatable Designs Derivable From Those of First Order. *Ann. Math. Statist.*, 31(4), 838-864. doi:10.1214/aoms/1177705661
- Brooks, C. A., & Cramer, S. M. (1992). Steric mass-action ion exchange: Displacement profiles and induced salt gradients. *AIChE Journal*, 38(12), 1969-1978. doi:10.1002/aic.690381212
- Bussamra, B. C., Sietaram, D., Verheijen, P., Mussatto, S. I., da Costa, A. C., van der Wielen, L., & Ottens, M. (2021). A critical assessment of the Flory-Huggins (FH) theory to predict aqueous two-phase behaviour. *Separation and Purification Technology*, 255, 117636. doi:<https://doi.org/10.1016/j.seppur.2020.117636>
- Buyel, J. F., Woo, J. A., Cramer, S. M., & Fischer, R. (2013). The use of quantitative structure–activity relationship models to develop optimized processes for the removal of tobacco host cell proteins during biopharmaceutical production. *Journal of Chromatography A*, 1322, 18-28. doi:<https://doi.org/10.1016/j.chroma.2013.10.076>
- Carta, G., & Jungbauer, A. (2010). *Protein Chromatography: Process Development and Scale-up*.
- Carvalho, S. B., Peixoto, C., Carrondo, M. J. T., & Silva, R. J. S. (2021). Downstream processing for influenza vaccines and candidates: An update. *Biotechnology and Bioengineering*, n/a(n/a). doi:<https://doi.org/10.1002/bit.27803>
- Chen, J., & Cramer, S. M. (2007). Protein adsorption isotherm behavior in hydrophobic interaction chromatography. *Journal of Chromatography A*, 1165(1), 67-77. doi:<https://doi.org/10.1016/j.chroma.2007.07.038>
- Chiang, M.-J., Pagkaliwangan, M., Lute, S., Bolton, G., Brorson, K., & Schofield, M. (2019). Validation and optimization of viral clearance in a downstream continuous chromatography setting. *Biotechnology and Bioengineering*, 116(9), 2292-2302. doi:<https://doi.org/10.1002/bit.27023>
- Close, E. J., Salm, J. R., Bracewell, D. G., & Sorensen, E. (2014). A model based approach for identifying robust operating conditions for industrial chromatography with process variability. *Chemical Engineering Science*, 116, 284-295. doi:<https://doi.org/10.1016/j.ces.2014.03.010>
- Czitrom, V. (1999). One-Factor-at-a-Time versus Designed Experiments. *The American Statistician*, 53(2), 126-131. doi:10.1080/00031305.1999.10474445
- Dismer, F., & Hubbuch, J. (2010). 3D structure-based protein retention prediction for ion-exchange chromatography. *Journal of Chromatography A*, 1217(8), 1343-1353. doi:<https://doi.org/10.1016/j.chroma.2009.12.061>
- Ehreth, J. (2003). The global value of vaccination. *Vaccine*, 21(7), 596-600. doi:[https://doi.org/10.1016/S0264-410X\(02\)00623-0](https://doi.org/10.1016/S0264-410X(02)00623-0)
- Eon-Duval, A., Gumbs, K., & Ellett, C. (2003). Precipitation of RNA impurities with high salt in a plasmid DNA purification process: Use of experimental design to determine reaction conditions. *Biotechnology and Bioengineering*, 83(5), 544-553. doi:<https://doi.org/10.1002/bit.10704>
- FDA. (2004). PAT Guidance for Industry - A Framework for innovative Pharmaceutical Development, Manufacturing and Quality Assurance. Retrieved from www.fda.gov/regulatory-information/search-fda-guidance-documents/pat-framework-innovative-pharmaceutical-development-manufacturing-and-quality-assurance

- Felinger, A., & Guiochon, G. (2004). Comparison of the Kinetic Models of Linear Chromatography. *Chromatographia*, 60(1), S175-S180. doi:10.1365/s10337-004-0288-7
- Ferreira, S. L. C., Bruns, R. E., da Silva, E. G. P., dos Santos, W. N. L., Quintella, C. M., David, J. M., de Andrade, J. B., Breitzkreitz, M. C., Jardim, I. C. S. F., & Neto, B. B. (2007). Statistical designs and response surface techniques for the optimization of chromatographic systems. *Journal of Chromatography A*, 1158(1), 2-14. doi:<https://doi.org/10.1016/j.chroma.2007.03.051>
- Gomis-Fons, J., Schwarz, H., Zhang, L., Andersson, N., Nilsson, B., Castan, A., Solbrand, A., Stevenson, J., & Chotteau, V. (2020). Model-based design and control of a small-scale integrated continuous end-to-end mAb platform. *Biotechnology Progress*, 36(4), e2995. doi:<https://doi.org/10.1002/btpr.2995>
- Hagel, L., Jagschies, G., & Sofer, G. (2008). *Handbook of Process Chromatography, Development, Manufacturing, Validation and Economics*.
- Hagen, A., Aunins, J., DePhillips, P., Oswald, C. B., Hennessey Jr, J. P., Lewis, J., Armstrong, M., Oliver, C., Orella, C., Buckland, B., & Sitrin, R. (2000). Development, preparation, and testing of VAQTA®, a highly purified hepatitis A vaccine. *Bioprocess Engineering*, 23(5), 439-449. doi:10.1007/s004499900157
- Hahn, T., Huuk, T., Heuveline, V., & Hubbuch, J. (2015). Simulating and Optimizing Preparative Protein Chromatography with ChromX. *Journal of Chemical Education*, 92(9), 1497-1502. doi:10.1021/ed500854a
- Hamidi, A., Kreeftenberg, H., van der Pol, L., Ghimire, S., van der Wielen, L. A. M., & Ottens, M. (2016). Process Development of a New Haemophilus influenzae Type b Conjugate Vaccine and the Use of Mathematical Modeling to Identify Process Optimization Possibilities. *Biotechnology Progress*, 32(3), 568-580. doi:10.1002/btpr.2235
- Hanke, A. T., & Ottens, M. (2014). Purifying biopharmaceuticals: knowledge-based chromatographic process development. *Trends in Biotechnology*, 32(4), 210-220. doi:10.1016/j.tibtech.2014.02.001
- Hanke, A. T., Tsintavi, E., Vazquez, M. D. R., van der Wielen, L. A. M., Verhaert, P. D. E. M., Eppink, M. H. M., van de Sandt, E. J. A. X., & Ottens, M. (2016). 3D-liquid chromatography as a complex mixture characterization tool for knowledge-based downstream process development. *Biotechnology Progress*, 32(5), 1283-1291. doi:10.1002/btpr.2320
- Havel, J., Madden, J. E., & Haddad, P. R. (1999). Prediction of retention times for anions in ion chromatography using Artificial Neural Networks. *Chromatographia*, 49(9), 481. doi:10.1007/BF02467746
- Hibbert, D. B. (2012). Experimental design in chromatography: A tutorial review. *Journal of Chromatography B*, 910, 2-13. doi:<https://doi.org/10.1016/j.jchromb.2012.01.020>
- Huuk, T. C., Hahn, T., Osberghaus, A., & Hubbuch, J. (2014). Model-based integrated optimization and evaluation of a multi-step ion exchange chromatography. *Separation and Purification Technology*, 136, 207-222. doi:10.1016/j.seppur.2014.09.012
- ICH. (2009). *ICH Harmonised Tripartite Guideline: Pharmaceutical Development Q8 (R2)*. Paper presented at the ICH.
- Jones, M., Palackal, N., Wang, F., Gaza-Bulseco, G., Hurkmans, K., Zhao, Y., Chitikila, C., Clavier, S., Liu, S., Menesale, E., Schonenbach, N. S., Sharma, S., Valax, P., Waerner, T., Zhang, L., & Connolly, T. (2021). "HIGH-RISK" HOST CELL PROTEINS (HCPs): A MULTI-COMPANY COLLABORATIVE VIEW. *Biotechnology and Bioengineering*, n/a(n/a). doi:<https://doi.org/10.1002/bit.27808>
- Josefsberg, J. O., & Buckland, B. (2012). Vaccine process technology. *Biotechnology and Bioengineering*, 109(6), 1443-1460. doi:10.1002/bit.24493
- Kalbfuss-Zimmermann, B., & Reichl, U. (2014). Viral Vaccines Purification. *Vaccine Development and Manufacturing*, 97-180. doi:<https://doi.org/10.1002/9781118870914.ch5>

- Kittelmann, J., Lang, K. M. H., Ottens, M., & Hubbuch, J. (2017a). Orientation of monoclonal antibodies in ion-exchange chromatography: A predictive quantitative structure–activity relationship modeling approach. *Journal of Chromatography A*, 1510, 33-39. doi:<https://doi.org/10.1016/j.chroma.2017.06.047>
- Kittelmann, J., Lang, K. M. H., Ottens, M., & Hubbuch, J. (2017b). An orientation sensitive approach in biomolecule interaction quantitative structure–activity relationship modeling and its application in ion-exchange chromatography. *Journal of Chromatography A*, 1482, 48-56. doi:<https://doi.org/10.1016/j.chroma.2016.12.065>
- Kleijnen, J. P. C. (2015). Response Surface Methodology. In *Handbook of Simulation Optimization* (pp. 81-104): New York, NY : Springer New York : Springer.
- Konstantinidis, S., Heldin, E., Chhatre, S., Velayudhan, A., & Titchener-Hooker, N. (2012). Strategic assay deployment as a method for countering analytical bottlenecks in high throughput process development: Case studies in ion exchange chromatography. *Biotechnology Progress*, 28(5), 1292-1302. doi:<https://doi.org/10.1002/btpr.1591>
- Krammer, F. (2020). SARS-CoV-2 vaccines in development. *Nature*, 586(7830), 516-527. doi:10.1038/s41586-020-2798-3
- Krippel, M., Dürauer, A., & Duerkop, M. (2020). Hybrid modeling of cross-flow filtration: Predicting the flux evolution and duration of ultrafiltration processes. *Separation and Purification Technology*, 248, 117064. doi:<https://doi.org/10.1016/j.seppur.2020.117064>
- Kumar, V., Bhalla, A., & Rathore, A. S. (2014). Design of experiments applications in bioprocessing: Concepts and approach. *Biotechnology Progress*, 30(1), 86-99. doi:10.1002/btpr.1821
- Lacki, K. M. (2012). High-throughput process development of chromatography steps: advantages and limitations of different formats used. *Biotechnol J*, 7(10), 1192-1202. doi:10.1002/biot.201100475
- Lacki, K. M. (2014). High throughput process development in biomanufacturing. *Current Opinion in Chemical Engineering*, 6, 25-32. doi:10.1016/j.coche.2014.08.004
- Ladd Effio, C., Hahn, T., Seiler, J., Oelmeier, S. A., Asen, I., Silberer, C., Villain, L., & Hubbuch, J. (2016). Modeling and simulation of anion-exchange membrane chromatography for purification of Sf9 insect cell-derived virus-like particles. *Journal of Chromatography A*, 1429, 142-154. doi:<https://doi.org/10.1016/j.chroma.2015.12.006>
- Langmuir, I. (1916). THE CONSTITUTION AND FUNDAMENTAL PROPERTIES OF SOLIDS AND LIQUIDS. PART I. SOLIDS. *Journal of the American Chemical Society*, 38(11), 2221-2295. doi:10.1021/ja02268a002
- Leser, E. W., & Asenjo, J. A. (1992). Rational design of purification processes for recombinant proteins. *Journal of Chromatography B: Biomedical Sciences and Applications*, 584(1), 43-57. doi:[https://doi.org/10.1016/0378-4347\(92\)80008-E](https://doi.org/10.1016/0378-4347(92)80008-E)
- Leweke, S., & von Lieres, E. (2018). Chromatography Analysis and Design Toolkit (CADET). *Computers & Chemical Engineering*, 113, 274-294. doi:<https://doi.org/10.1016/j.compchemeng.2018.02.025>
- Lienqueo, M. E., & Asenjo, J. A. (2000). Use of expert systems for the synthesis of downstream protein processes. *Computers & Chemical Engineering*, 24(9), 2339-2350. doi:[https://doi.org/10.1016/S0098-1354\(00\)00590-1](https://doi.org/10.1016/S0098-1354(00)00590-1)
- Liu, S., Gerontas, S., Gruber, D., Turner, R., Titchener-Hooker, N. J., & Papageorgiou, L. G. (2017). Optimization-based framework for resin selection strategies in biopharmaceutical purification process development. *Biotechnology Progress*, 33(4), 1116-1126. doi:10.1002/btpr.2479
- Ljung, L., & Glad, T. (1994). *Modeling of dynamic systems*: Englewood Cliffs (N.J.) : Prentice-Hall.
- Löfgren, A., Yamanee-Nolin, M., Tallvod, S., Fons, J. G., Andersson, N., & Nilsson, B. (2019). Optimization of integrated chromatography sequences for purification of biopharmaceuticals. *Biotechnology Progress*, 35(6), e2871. doi:<https://doi.org/10.1002/btpr.2871>
- Lovrien, R. E., & Matulis, D. (1997). Selective Precipitation of Proteins. *Current Protocols in Protein Science*, 7(1), 4.5.1-4.5.36. doi:<https://doi.org/10.1002/0471140864.ps0405s07>

- Marengo, E., Gennaro, M. C., & Angelino, S. (1998). Neural network and experimental design to investigate the effect of five factors in ion-interaction high-performance liquid chromatography. *Journal of Chromatography A*, 799(1), 47-55. doi:[https://doi.org/10.1016/S0021-9673\(97\)01027-3](https://doi.org/10.1016/S0021-9673(97)01027-3)
- Markarian, J. (2018). Industry 4.0 in Biopharmaceutical Manufacturing. *Biopharm International*, 31
- Meyer, K., Leweke, S., von Lieres, E., Huusom, J. K., & Abildskov, J. (2020). ChromaTech: A discontinuous Galerkin spectral element simulator for preparative liquid chromatography. *Computers & Chemical Engineering*, 141, 107012. doi:<https://doi.org/10.1016/j.compchemeng.2020.107012>
- Moreno-González, M., Girish, V., Keulen, D., Wijngaard, H., Lauteslager, X., Ferreira, G., & Ottens, M. (2020). Recovery of sinapic acid from canola/rapeseed meal extracts by adsorption. *Food and Bioprocess Technology*, 120, 69-79. doi:<https://doi.org/10.1016/j.fbp.2019.12.002>
- Moreno-González, M., Keulen, D., Gomis-Fons, J., Gomez, G. L., Nilsson, B., & Ottens, M. (2021). Continuous adsorption in food industry: The recovery of sinapic acid from rapeseed meal extract. *Separation and Purification Technology*, 254, 117403. doi:<https://doi.org/10.1016/j.seppur.2020.117403>
- Muller, P.-A., Fondement, F., Baudry, B., & Combemale, B. (2012). Modeling modeling modeling. *Software & Systems Modeling*, 11(3), 347-359. doi:10.1007/s10270-010-0172-x
- Nagrath, D., Messac, A., B, W. B., & S, M. C. (2004). A Hybrid Model Framework for the Optimization of Preparative Chromatographic Processes. *Biotechnology Progress*, 20(1), 162-178. doi:10.1021/bp034026g
- Nfor, B. K., Ahamed, T., Pinkse, M. W. H., van der Wielen, L. A. M., Verhaert, P. D. E. M., van Dedem, G. W. K., Eppink, M. H. M., van de Sandt, E. J. A. X., & Ottens, M. (2012a). Multi-dimensional fractionation and characterization of crude protein mixtures: Toward establishment of a database of protein purification process development parameters. *Biotechnology and Bioengineering*, 109(12), 3070-3083. doi:10.1002/bit.24576
- Nfor, B. K., Ahamed, T., van Dedem, G. W. K., van der Wielen, L. A. M., van de Sandt, E. J. A. X., Eppink, M. H. M., & Ottens, M. (2008). Design strategies for integrated protein purification processes: challenges, progress and outlook. *Journal of Chemical Technology and Biotechnology*, 83(2), 124-132. doi:10.1002/jctb.1815
- Nfor, B. K., Ahamed, T., van Dedem, G. W. K., Verhaert, P. D. E. M., van der Wielen, L. A. M., Eppink, M. H. M., van de Sandt, E. J. A. X., & Ottens, M. (2013). Model-based rational methodology for protein purification process synthesis. *Chemical Engineering Science*, 89, 185-195. doi:10.1016/j.ces.2012.11.034
- Nfor, B. K., Hylkema, N. N., Wiedhaup, K. R., Verhaert, P. D. E. M., van der Wielen, L. A. M., & Ottens, M. (2011a). High-throughput protein precipitation and hydrophobic interaction chromatography: Salt effects and thermodynamic interrelation. *Journal of Chromatography A*, 1218(49), 8958-8973. doi:10.1016/j.chroma.2011.08.016
- Nfor, B. K., Noverraz, M., Chilamkurthi, S., Verhaert, P. D. E. M., van der Wielen, L. A. M., & Ottens, M. (2010). High-throughput isotherm determination and thermodynamic modeling of protein adsorption on mixed mode adsorbents. *Journal of Chromatography A*, 1217(44), 6829-6850. doi:10.1016/j.chroma.2010.07.069
- Nfor, B. K., Ripic, J., van der Padt, A., Jacobs, M., & Ottens, M. (2012b). Model-based high-throughput process development for chromatographic whey proteins separation. *Biotechnology Journal*, 7(10), 1221-1232. doi:10.1002/biot.201200191
- Nfor, B. K., Verhaert, P. D. E. M., van der Wielen, L. A. M., Hubbuch, J., & Ottens, M. (2009). Rational and systematic protein purification process development: the next generation. *Trends in Biotechnology*, 27(12), 673-679. doi:10.1016/j.tibtech.2009.09.002
- Nfor, B. K., Zuluaga, D. S., Verheijen, P. J. T., Verhaert, P. D. E. M., van der Wielen, L. A. M., & Ottens, M. (2011b). Model-based rational strategy for chromatographic resin selection. *Biotechnology Progress*, 27(6), 1629-1643. doi:<https://doi.org/10.1002/btpr.691>

- Nikita, S., Tiwari, A., Sonawat, D., Kodamana, H., & Rathore, A. S. (2020). Reinforcement Learning based Optimization of Process Chromatography for Continuous Processing of Biopharmaceuticals. *Chemical Engineering Science*, 116171. doi:<https://doi.org/10.1016/j.ces.2020.116171>
- Oelmeier, S. A., Dimer, F., & Hubbuch, J. (2011). Application of an aqueous two-phase systems high-throughput screening method to evaluate mAb HCP separation. *Biotechnology and Bioengineering*, 108(1), 69-81. doi:<https://doi.org/10.1002/bit.22900>
- Osberghaus, A., Hepbildikler, S., Nath, S., Haindl, M., von Lieres, E., & Hubbuch, J. (2012). Determination of parameters for the steric mass action model—A comparison between two approaches. *Journal of Chromatography A*, 1233, 54-65. doi:<https://doi.org/10.1016/j.chroma.2012.02.004>
- Otero, B., Degerman, M., Hansen, T. B., Hansen, E. B., & Nilsson, B. (2014). Model-based design and integration of a two-step biopharmaceutical production process. *Bioprocess and Biosystems Engineering*, 37(10), 1989-1996. doi:10.1007/s00449-014-1174-9
- Papathanasiou, M. M., Steinebach, F., Morbidelli, M., Mantalaris, A., & Pistikopoulos, E. N. (2017). Intelligent, model-based control towards the intensification of downstream processes. *Computers & Chemical Engineering*, 105, 173-184. doi:<https://doi.org/10.1016/j.compchemeng.2017.01.005>
- Parimal, S., Garde, S., & Cramer, S. M. (2015). Interactions of Multimodal Ligands with Proteins: Insights into Selectivity Using Molecular Dynamics Simulations. *Langmuir*, 31(27), 7512-7523. doi:10.1021/acs.langmuir.5b00236
- Pirrung, S. M., Berends, C., Backx, A. H., van Beckhoven, R. F. W. C., Eppink, M. H. M., & Ottens, M. (2019). Model-based optimization of integrated purification sequences for biopharmaceuticals. *Chemical Engineering Science: X*, 3, 100025. doi:<https://doi.org/10.1016/j.cesx.2019.100025>
- Pirrung, S. M., da Cruz, D. P., Hanke, A. T., Berends, C., van Beckhoven, R. F. W. C., Eppink, M. H. M., & Ottens, M. (2018). Chromatographic Parameter Determination for Complex Biological Feedstocks. *Biotechnology Progress*, 34(4), 1006-1018. doi:10.1002/btpr.2642
- Pirrung, S. M., & Ottens, M. (2017). High Throughput Process Development. In A. Staby, A. S. Rathore, & S. Ahuja (Eds.), *Preparative Chromatography for Separation of Proteins* (pp. 269 - 292): John Wiley & Sons, Inc.
- Pirrung, S. M., van der Wielen, L. A. M., van Beckhoven, R. F. W. C., van de Sandt, E. J. A. X., Eppink, M. H. M., & Ottens, M. (2017). Optimization of biopharmaceutical downstream processes supported by mechanistic models and artificial neural networks. *Biotechnology Progress*, 33(3), 696-707. doi:10.1002/btpr.2435
- Pollard, A. J., & Bijker, E. M. (2021). A guide to vaccinology: from basic principles to new developments. *Nature Reviews Immunology*, 21(2), 83-100. doi:10.1038/s41577-020-00479-7
- Portela, R. M. C., Varsakelis, C., Richelle, A., Giannelos, N., Pence, J., Desso, S., & von Stosch, M. (2020). When Is an In Silico Representation a Digital Twin? A Biopharmaceutical Industry Approach to the Digital Twin Concept. In *Digital Twins* (pp. 35-55). Berlin, Heidelberg: Springer Berlin Heidelberg.
- Psichogios, D. C., & Ungar, L. H. (1992). A hybrid neural network-first principles approach to process modeling. *AIChE Journal*, 38(10), 1499-1511. doi:10.1002/aic.690381003
- Rappuoli, R. (2007). Bridging the knowledge gaps in vaccine design. *Nature Biotechnology*, 25(12), 1361-1366. doi:10.1038/nbt1207-1361
- Rathore, A. S. (2009). Roadmap for implementation of quality by design (QbD) for biotechnology products. *Trends in Biotechnology*, 27(9), 546-553. doi:10.1016/j.tibtech.2009.06.006
- Rathore, A. S. (2016). Quality by Design (QbD)-Based Process Development for Purification of a Biotherapeutic. *Trends in Biotechnology*, 34(5), 358-370. doi:10.1016/j.tibtech.2016.01.003
- Rathore, A. S., Kumar, D., & Kateja, N. (2018). Recent developments in chromatographic purification of biopharmaceuticals. *Biotechnology Letters*, 40(6), 895-905. doi:10.1007/s10529-018-2552-1

- Rathore, A. S., Mittal, S., Pathak, M., & Arora, A. (2014). Guidance for performing multivariate data analysis of bioprocessing data: Pitfalls and recommendations. *Biotechnology Progress*, 30(4), 967-973. doi:<https://doi.org/10.1002/btpr.1922>
- Rege, K., Pepsin, M., Falcon, B., Steele, L., & Heng, M. (2006). High-throughput process development for recombinant protein purification. *Biotechnology and Bioengineering*, 93(4), 618-630. doi:<https://doi.org/10.1002/bit.20702>
- Rodrigues, C. M. C., & Plotkin, S. A. (2020). Impact of Vaccines; Health, Economic and Social Perspectives. *Frontiers in Microbiology*, 11(1526). doi:10.3389/fmicb.2020.01526
- Ruthven, D. M. (1984). *Principles of adsorption and adsorption processes*. New York: Wiley.
- Saleh, D., Wang, G., Müller, B., Rischawy, F., Kluters, S., Studts, J., & Hubbuch, J. (2020). Straightforward method for calibration of mechanistic cation exchange chromatography models for industrial applications. *Biotechnology Progress*, n/a(n/a), e2984. doi:10.1002/btpr.2984
- São Pedro, M. N., Silva, T. C., Patil, R., & Ottens, M. (2021). White paper on high-throughput process development for integrated continuous biomanufacturing. *Biotechnology and Bioengineering*, n/a(n/a). doi:<https://doi.org/10.1002/bit.27757>
- Schmölder, J., & Kaspereit, M. (2020). A Modular Framework for the Modelling and Optimization of Advanced Chromatographic Processes. *Processes*, 8(1), 65. Retrieved from <https://doi.org/10.3390/pr8010065>
- Sciences, G. H. L. (2012). *Recombinant Protein Purification Handbook, Principles and Methods*.
- Selic, B. (2003). The pragmatics of model-driven development. *IEEE Software*, 20(5), 19-25. doi:10.1109/MS.2003.1231146
- Shekhawat, L. K., Chandak, M., & Rathore, A. S. (2017). Mechanistic modeling of hydrophobic interaction chromatography for monoclonal antibody purification: process optimization in the quality by design paradigm. *Journal of Chemical Technology and Biotechnology*, 92(10), 2527-2537. doi:10.1002/jctb.5324
- Shukla, A. A., Hubbard, B., Tressel, T., Guhan, S., & Low, D. (2007). Downstream processing of monoclonal antibodies—Application of platform approaches. *Journal of Chromatography B*, 848(1), 28-39. doi:<https://doi.org/10.1016/j.jchromb.2006.09.026>
- Shukla, A. A., & Thommes, J. (2010). Recent advances in large-scale production of monoclonal antibodies and related proteins. *Trends in Biotechnology*, 28(5), 253-261. doi:10.1016/j.tibtech.2010.02.001
- Siirola, J. J. (1996). Industrial Applications of Chemical Process Synthesis. In J. L. Anderson (Ed.), *Advances in Chemical Engineering* (Vol. 23, pp. 1-62): Academic Press.
- Singh, N., & Herzer, S. (2018). Downstream Processing Technologies/Capturing and Final Purification. In B. Kiss, U. Gottschalk, & M. Pohlscheidt (Eds.), *New Bioprocessing Strategies: Development and Manufacturing of Recombinant Antibodies and Proteins* (pp. 115-178). Cham: Springer International Publishing.
- Solle, D., Hitzmann, B., Herwig, C., Pereira Remelhe, M., Ulonska, S., Wuerth, L., Prata, A., & Steckenreiter, T. (2017). Between the Poles of Data-Driven and Mechanistic Modeling for Process Operation. *Chemie Ingenieur Technik*, 89(5), 542-561. doi:<https://doi.org/10.1002/cite.201600175>
- Stamatis, C., Goldrick, S., Gruber, D., Turner, R., Titchener-Hooker, N. J., & Farid, S. S. (2019). High throughput process development workflow with advanced decision-support for antibody purification. *Journal of Chromatography A*, 1596, 104-116. doi:10.1016/j.chroma.2019.03.005
- Susanto, A., Knieps-Grunhagen, E., von Lieres, E., & Hubbuch, J. (2008). High Throughput Screening for the Design and Optimization of Chromatographic Processes: Assessment of Model Parameter Determination from High Throughput Compatible Data. *Chemical Engineering & Technology*, 31(12), 1846-1855. doi:10.1002/ceat.200800457

- Touelle, M., Uzel, A., Depoisier, J.-F., & Gantier, R. (2011). Designing new monoclonal antibody purification processes using mixed-mode chromatography sorbents. *Journal of Chromatography B*, 879(13), 836-843. doi:<https://doi.org/10.1016/j.jchromb.2011.02.047>
- Vogg, S., Müller-Späth, T., & Morbidelli, M. (2020). Design space and robustness analysis of batch and counter-current frontal chromatography processes for the removal of antibody aggregates. *Journal of Chromatography A*, 1619, 460943. doi:<https://doi.org/10.1016/j.chroma.2020.460943>
- von Stosch, M., Oliveira, R., Peres, J., & de Azevedo, S. F. (2014). Hybrid semi-parametric modeling in process systems engineering: Past, present and future. *Computers & Chemical Engineering*, 60, 86-101. doi:10.1016/j.compchemeng.2013.08.008
- Wang, G., Briskot, T., Hahn, T., Baumann, P., & Hubbuch, J. (2017). Estimation of adsorption isotherm and mass transfer parameters in protein chromatography using artificial neural networks. *Journal of Chromatography A*, 1487, 211-217. doi:10.1016/j.chroma.2017.01.068
- Weigel, T., Soliman, R., Wolff, M. W., & Reichl, U. (2019). Hydrophobic-interaction chromatography for purification of influenza A and B virus. *Journal of Chromatography B*, 1117, 103-117. doi:<https://doi.org/10.1016/j.jchromb.2019.03.037>
- Welsh, J. P., Petroff, M. G., Rowicki, P., Bao, H., Linden, T., Roush, D. J., & Pollard, J. M. (2014). A practical strategy for using miniature chromatography columns in a standardized high-throughput workflow for purification development of monoclonal antibodies. *Biotechnology Progress*, 30(3), 626-635. doi:<https://doi.org/10.1002/btpr.1905>
- Wen, E. P., Ellis, R., & Pujar, N. S. (2014). *Vaccine Development and Manufacturing*: Wiley.
- Westerberg, K., Borg, N., Andersson, N., & Nilsson, B. (2012). Supporting Design and Control of a Reversed-Phase Chromatography Step by Mechanistic Modeling. *Chemical Engineering & Technology*, 35(1), 169-175. doi:<https://doi.org/10.1002/ceat.201000505>
- Wiendahl, M., Schulze Wierling, P., Nielsen, J., Fomsgaard Christensen, D., Krarup, J., Staby, A., & Hubbuch, J. (2008). High Throughput Screening for the Design and Optimization of Chromatographic Processes – Miniaturization, Automation and Parallelization of Breakthrough and Elution Studies. *Chemical Engineering & Technology*, 31(6), 893-903. doi:10.1002/ceat.200800167
- Woo, J., Parimal, S., Brown, M. R., Heden, R., & Cramer, S. M. (2015). The effect of geometrical presentation of multimodal cation-exchange ligands on selective recognition of hydrophobic regions on protein surfaces. *Journal of Chromatography A*, 1412, 33-42. doi:<https://doi.org/10.1016/j.chroma.2015.07.072>
- Yang, Y.-p., & D'Amore, T. (2014). Protein Subunit Vaccine Purification. In *Vaccine Development and Manufacturing* (pp. 181-216).
- Yu, L. X. (2008). Pharmaceutical Quality by Design: Product and Process Development, Understanding, and Control. *Pharmaceutical Research*, 25(4), 781-791. doi:10.1007/s11095-007-9511-1
- Zhao, M., Vandersluis, M., Stout, J., Haupts, U., Sanders, M., & Jacquemart, R. (2019). Affinity chromatography for vaccines manufacturing: Finally ready for prime time? *Vaccine*, 37(36), 5491-5503. doi:<https://doi.org/10.1016/j.vaccine.2018.02.090>
- Zimmermann, S., Gretzinger, S., Schwab, M.-L., Scheeder, C., Zimmermann, P. K., Oelmeier, S. A., Gottwald, E., Bogsnes, A., Hansson, M., Staby, A., & Hubbuch, J. (2016). High-throughput downstream process development for cell-based products using aqueous two-phase systems. *Journal of Chromatography A*, 1464, 1-11. doi:<https://doi.org/10.1016/j.chroma.2016.08.025>

List of Tables

Table 1. Overview of the main advantages and disadvantageous of different modeling approaches.

	Advantages	Shortcomings
Data-driven models	- Requires no or little process understanding in advance - Takes less effort/time to develop the model - Easy to use and understand	- Only valid in a predefined measured region - Extrapolation generally not applicable - Parameters have often no physical meaning
Mechanistic models	- Allows extrapolation and exploration of conditions beyond measured results - Acquires process understanding - Parameters have a physical meaning	- Requires process understanding in advance - Complex to develop and hence time and effort
Hybrid models	- Eliminate drawbacks of certain modeling approaches - Improved model accuracy and extrapolation properties	- Requires more data to make another model - Requires additional effort/time to develop the model

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902 Table 2. Overview of HTPD approaches which have been applied in industry and/or academia using HT and/or modelling techniques.

Purpose of re- search	Experimental Method			Modelling method	Application, unit op- eration and stage of development	Ref
	HT	DoE	Lab-scale	Empirical or Mechanistic		
Defining charac- terization/oper- ating space	Robocolumns for condition screening	Full factorial DoE, varying two process variables, pH and buffer molarity.	None	Empirical: Regression analysis by Least square fitting and optimization using contour profiler.	- GCSF - IEX - Early	(Bhambure & Rathore, 2013)
	Batch adsorption experiments for isotherms and Robocolumns breakthrough experiments (DBC)	None	Lab-scale columns, validation experiments	Mechanistic: Regression of partitioning coefficient and maximum binding capacity, Langmuir isotherm model.	- mAb - IEX, MMC - Early	(Welsh et al., 2014)
Resin selection and salt concentration	Batch adsorption experiments for condition screening	None	Lab-scale column, breakthrough experiments (DBC – 10%)	None	- Virus, influenza A, B - HIC - Early	(Weigel et al., 2019)
Design bind-and-elute membrane process	Batch binding and buffer screenings	Buffer screenings	Lab-scale column, membrane characterization, breakthrough experiments	Mechanistic: General rate model for radial flow chromatography Regression and chromatogram fitting for estimating isotherm and model parameters	- Virus Like Particles - AEX, membrane chromatography - Early and late	(Ladd Effio et al., 2016)

Resin selection, optimization and defining operation window	Robocolumns for resin screening and optimization screenings	Definitive screening designs for resin screening Central composite designs for optimization screenings	Lab-scale column for model verification	Empirical: multivariate data analysis and usage of multi-criteria decision-making techniques. Process parameter optimization and Robustness analysis	- Highly aggregate antibody solution. - CEX - Early	(Stamatis et al., 2019)
Resin selection, multiple unit optimization	Robocolumns, bind-elute mode, resin and operating condition screening	None	None	Empirical: Multi-objective mixed integer nonlinear programming model. Adopted ϵ -constraint method solved by Dinkelbach's algorithm	Recombinant Fc Fusion protein - CEX - MMC - Early	(Liu et al., 2017)
Flowsheet optimization, resin selection, design of process	Batch adsorption experiments (HT) for isotherm determination and resin selection.	None	Lab-scale column experiments for validation and acquisition of molecular properties	Mechanistic: Flowsheet optimization top-to-bottom approach using chromatographic mechanistic models including adsorption isotherm models	- mAb from hybridoma cell culture - CEX, AEX, HIC, SEC - 4-steps - Early and late	(Nfor et al., 2012a; Nfor et al., 2013; Nfor et al., 2011b)
	Robocolumns to determine isotherm parameters Batch-uptake experiments for determining maximum binding capacities	None	Lab-scale column experiments for validation	Mechanistic: Flowsheet optimization, global optimization along with ANN and Local optimization along with Mechanistic models, including isotherm models	- mAb IgG1 from CHO - CEX, MMC, HIC, UF/DF - 4-steps - Early and late	(Pirrung et al., 2019) (Pirrung et al., 2018)
	None	None	Lab-scale breakthrough column experiments.	Mechanistic: Flowsheet optimization using mechanistic models, including isotherm models.	- applied to three model proteins - CEX, AEX - 2-steps - Early and late	(Huuk et al., 2014)

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905 Table 3. Glossary

Abbreviation	Explanation
<i>AI</i>	Artificial intelligence
<i>AEX</i>	Anion exchange chromatography
<i>ANN</i>	Artificial neural networks
<i>ATPs</i>	Aqueous two-phase systems
<i>CQA</i>	Critical quality attributes
<i>CEX</i>	Cation exchange chromatography
<i>CHO</i>	Chinese hamster ovary cells
<i>QbD</i>	Quality by design
<i>DF</i>	Diafiltration
<i>DoE</i>	Design of experiments
<i>DSP</i>	Downstream processing
<i>GCSF</i>	Granulocyte colony stimulating factor
<i>HIC</i>	Hydrophobic interaction chromatography
<i>HT</i>	High throughput
<i>HTE / HTS</i>	High throughput experimentation / screening
<i>HTPD</i>	High throughput process design
<i>HCP</i>	Host cell proteins
<i>IEX</i>	Ion exchange chromatography
<i>LHS</i>	Liquid handling station
<i>mAbs</i>	Monoclonal antibodies
<i>MMC</i>	Mixed-mode chromatography
<i>OFAT</i>	One-factor-at-a-time
<i>PAT</i>	Process analytical techniques
<i>QSAR</i>	Quantitative-structure-activity-relationships
<i>RL</i>	Reinforcement learning
<i>RSM</i>	Response surface methodology
<i>SEC</i>	Size-exclusion chromatography
<i>UF</i>	Ultrafiltration
<i>VLP</i>	Virus-like particles

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Figure 1. Types of vaccines classified in whole pathogen, antigenic components of pathogen and nucleic acid vaccines (Pollard & Bijker, 2021; Rappuoli, 2007; Wen et al., 2014).

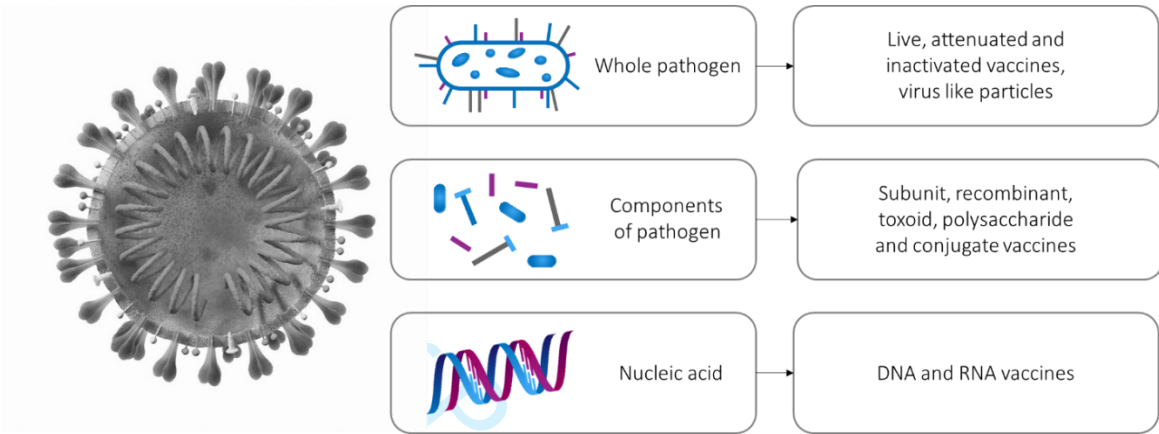


Figure 2. General process flowsheet for vaccines including the upstream and downstream part, from fermentation to the last formulating steps. The optional processing techniques for different types of vaccines are given below each unit operation.

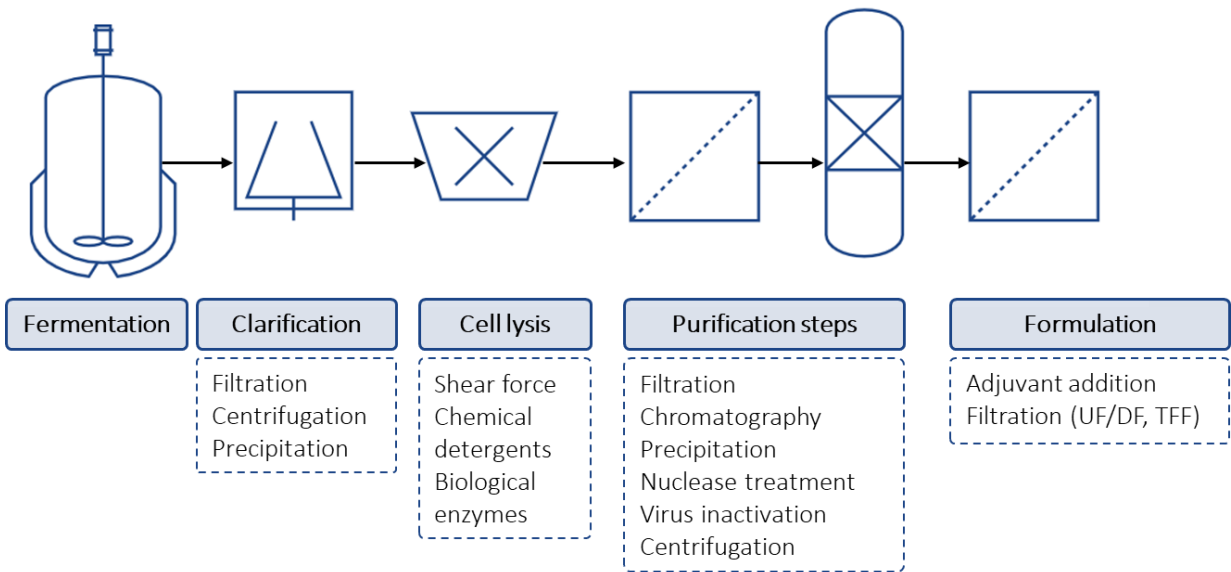


Figure 3. Overview of two different process development approaches. Left: Design of Experiments (DoE) approach most commonly applied within biopharmaceutical industry, in which the process design is based on performing experiments along with statistical methods. Right: Model-based process development approach employing targeted experiments which are intended as model input parameters. Optimization is performed using a validated model.

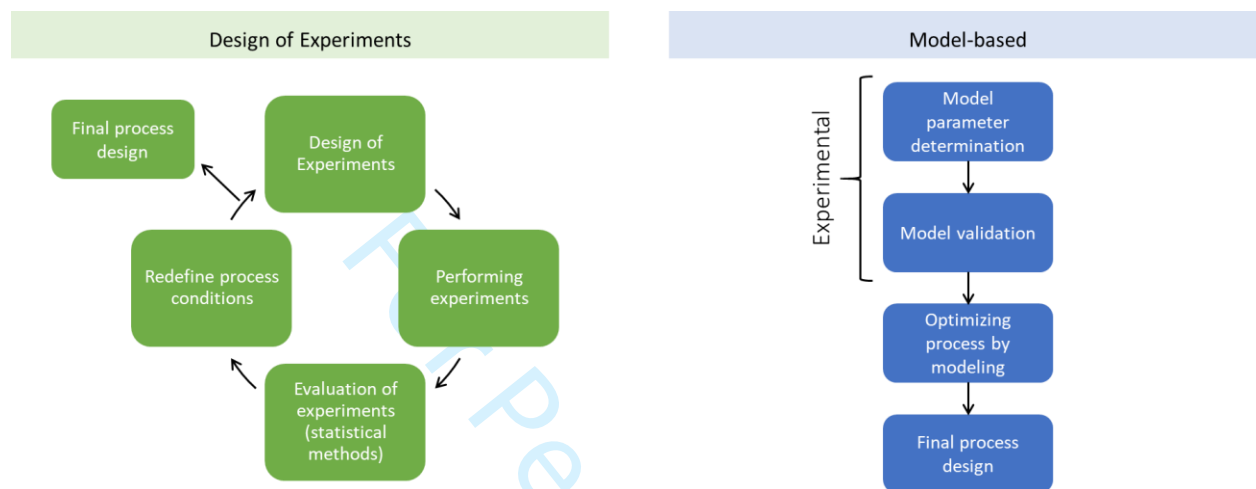
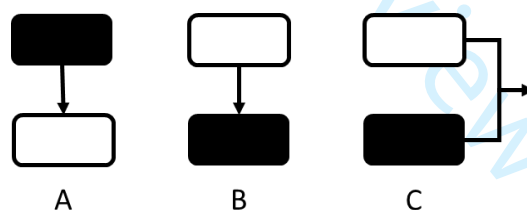


Figure 4. Hybrid modeling configurations, white-boxes represent mechanistic/first-principle models and black-boxes represent data-driven models (von Stosch et al., 2014).



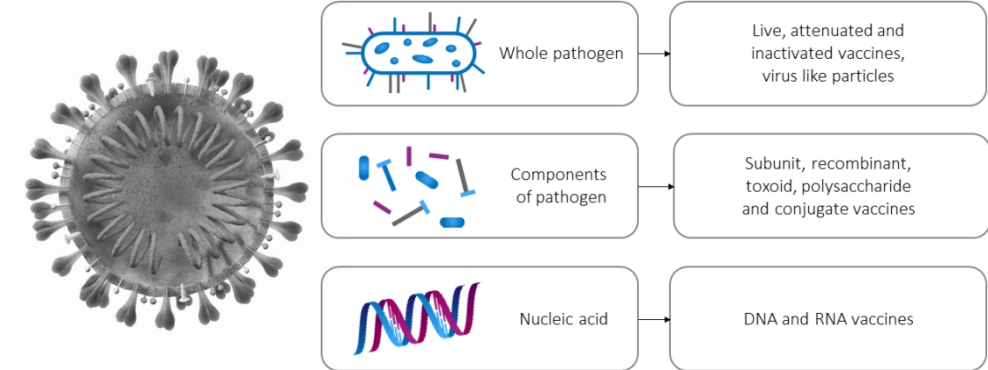


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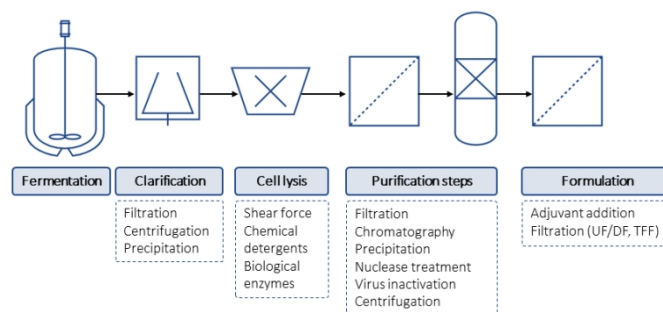


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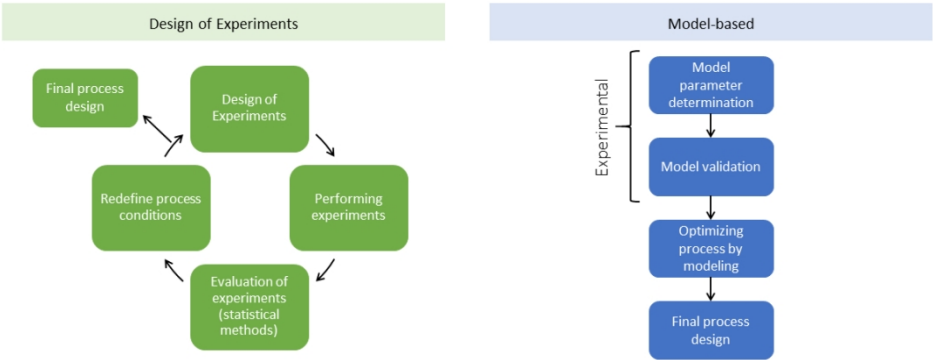


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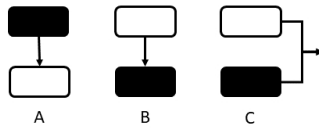


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